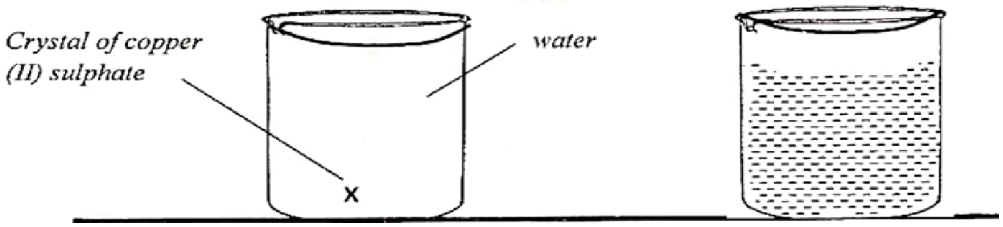
BE TRUE

CHEMISTRY DEPARTMENT
2022 MOCK EXAMINATION SETS

TORORO ARCHDIOSCESE EXAMINATION BOARD:

SECTION A

1. Figure 1.0 shows a coloured crystal of copper (II) sulphate is placed at the bottom of the beaker containing water. After two days, the colour had spread throughout the water.



(a) Explain these observations. (2 marks)

(b) Describe how you would prepare a pure sample of crystals of hydrated (II) sulphate starting with copper (II) oxide. Write an equation for the reaction. (3 marks)

2. The number of protons, neutrons and orbital electrons in particles A to F of unknown elements are given in the following table.

Particle	Proton	Neutrons	Electrons
A	8	8	10
B	8	10	8
C	11	12	11
D	13	14	10
E	14	14	14
F	18	22	18

(a) Choose from the table the letter(s) that represent the following descriptions. (3½ marks)

(i) It is a soft metal with low density.

(ii) It has a giant covalent structure similar to diamond.

(iii) It is an atom of a noble gas (inert gas)

(iv) A pair of isotopes.

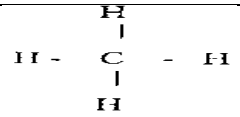
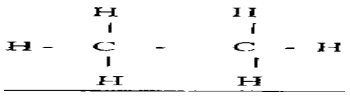
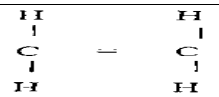
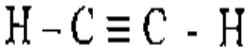
(v) Is a cation (positive ion)

(vi) It is a diatomic gas with molecules of the type X_3.

(vii) It is an anion (negative ion)

(b) Using outermost energy level electrons only, draw diagrams to show how elements B and C react to form a compound, use x and e to represent the electron (s) from B and C respectively.

3.

Compound	Structure formula
A	
B	
C	
D	

(a) (i) What general term is used for compounds, like those shown in the table, that contain only carbon and hydrogen atoms? (½ mark)

(ii) To which homologous series do A and B belongs? (½ mark)

(b) Which compound in the table,

(i) When burned in oxygen, produces a very high temperature flame used for cutting and welding metals. (½ mark)

(ii) Is produced by anaerobic bacteria decomposing animal waste? (½ mark)

(c) Why is the concentration of this compound in (b) (ii) in the atmosphere a cause for concern? (½ mark)

(d) (i) If a compound contain 80% carbon and 20% hydrogen and its relative molecular mass is 30, what is its molecular formula?

(ii) Give one test which distinguishes between this hydro carbon in d (i) and the corresponding hydrocarbon with two fewer hydrogen atoms in its molecules. (1½ marks)

4. Metals have characteristic physical properties such as good electrical and thermal conductivity.

(a) Give three other physical properties which are characteristics of metals. (1½ marks)

(b) The table below gives some observations about the reactions of four metals with water.

Meta	Observations
Zinc	Water does not attack zinc to any appreciable extent. Zinc at a red heat is attacked by steam with the formation of hydrogen
Magnesium	Reacts slowly with cold water.
Iron reacts with steam	Reacts with steam when red hot
Sodium	Reacts rapidly with cold water

Put these metals in the order of their decreasing reactivity with water (1 mark)

(c) The equation for the reaction of iron with steam is shown.

$$3Fe\ (s) + 4\ H_2O\ (g) \rightleftharpoons Fe_3O_4\ (s) + 4\ H_2\ (g)$$

Calculate the mass of tri iron tetra oxide, Fe_3O_4 that can be formed when 39.2g of iron reacts with excess steam. (2 marks)

(d) State one large scale use of iron. (½ mark)

5. An aqueous solution of ammonium nitrite, NH_4NO_2. Decomposes when heated gently according to the following equation.

$$NH_4NO_2\ (aq) \longrightarrow N_2\ (g) + 2\ H_2O\ (l)$$

(a) Describe how you could show that aqueous ammonium nitrite contains ammonium ions. (2 marks)

(b) A sample of 25.0 cm^3 of 0.50 mol/dm^3 aqueous ammonium nitrite is heated. Calculate the volume of nitrogen formed at room temperature and pressure. (2 marks)

(c) Ammonium nitrate NH_4NO_3, decomposes when heated in a similar way to ammonium nitrite. Identify the two products made in this reaction. (1 mark)

6. Dilute sulphuric acid has typical acid properties. An excess of it is added to test tubes W, X,Y,and Z, which contain these powdered substances:

W-copper (II) oxide. X-sodium hydroxide. Y- magnesium Z -sodium carbonate.

(a) (i) In which test tube (s) will you observe fissing (effervescence) (½ mark)

(ii) In which test tube does neutralization take place ? (½ mark)

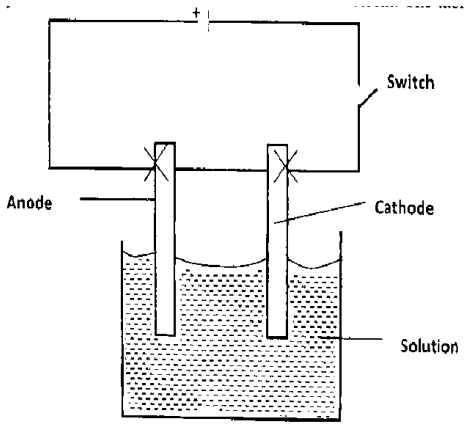
(iii) In which test tube will a blue solution form. (½ mark)

(b) Name the three (3) salts obtained after reaction. (1½ marks)

(c) Write ionic equations for: (i) the neutrlaization of an acid by an alkali. (1 mark)

(ii) the reaction between magnesium and dilute acid. (1 mark)

7. Six substances A to F were dissolved in water, and connected in turn into the circuit below, A represents an ammeter which is used to measure current. The table shows the results



Subsatnce	Current (amperes)	At cathode	At anode
A	O.8	Copper	Chlorine
B	1.0	Hydrogen	Chlorine
C	0.0	-	-
D	0.8	Copper	Oxygen
E	1.2	Hydrogen	Oxygen
F	0.7	Silver	Oxygen

(a) (i) Which solution conducts best? (½ mark)

(ii) Which solution is a non- electrolyte? (½ mark)

(b) Which solution could be: (3 marks)

(i) Copper (II) sulphate?
(ii) Silver nitrate?
(iii) Sodium hydroxide?
(iv) Copper (II) chloride?
(v) Ethanol (and or sugar)
(vi) Concentrated hydrochloric acid?

(c) Explain how the current is carried:

(i) within the electrolytes. (½ mark)
(ii) in the rest of the circuit. (½ mark)

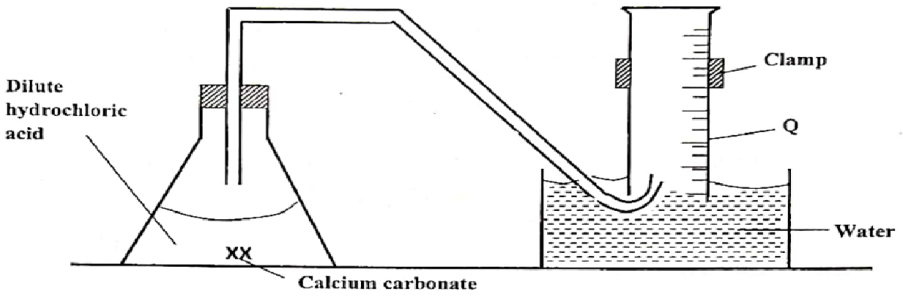
8. Name the catalyst used in the following processes. (3 ½ marks)

Process	Catalyst.
(a) Decomposition of hydrogen peroxide. $2H_2O_2(aq) \longrightarrow 2H_2O(l) + O_2(g)$	
(b) Haber process: for the manufacture of ammonia. $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$	
(c) Heating potassium chlorate $2KClO_3(s) \longrightarrow 2KCl(s) + 3O_2(g)$	
(d) Hydrogenation of unsaturated oils to form fats in manufacture of margarines. $C_4H_4(g) + H_2(g) \longrightarrow C_2H_6(g)$	
(e) The reaction between zinc granules and cold dilute sulphuric acid. $Zn(s) + H_2SO_4(aq) \longrightarrow ZnSO_4(l) + H_2(g)$	
(f) Contact process for the manufacture of sulphuric acid. $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$	
(g) Oxidation of ammonia in the manufacture of nitric acid. $4NH_3(g) + 5O_2(g) \longrightarrow 4NO\ (g) + 6H_2O(l)$	

9. State what would be observed and write the equations for the reaction that would take place when: (4 marks)

(a) Aqueous ammonium sulphate was heated with a solution of potassium hydroxide.
(b) An aqueous solution of hydrogen chloride reacted with lead (II) carbonate.

10. The apparatus shown is used to investigate the rate of reaction between calcium carbonate and dilute hydrochloric acid at 30^0C.



(a) Write an equation for the reaction. (1 mark)
(b) (i) Name the piece of apparatus labeled Q in the diagram. (½ mark)
(ii)Describe how this apparatus can be used to find the rate of reaction. (1 mark)
(c) The experiment is repeated at 20^0C. All other conditions are kept the same. How does the decrease in temperature affect the rate of reaction? (1 mark)

(d) The experiment is repeated using the same mass of different size pieces of calcium carbonate. All other conditions are kept the same. The sizes of the pieces of calcium carbonate are:

-Powder. -Large pieces of marble. -Small pieces of marble.

Complete the table by writing the sizes of the pieces in the first column. (1½ marks)

Size of the pieces of calcium carbonate	Initial rate of reaction in (cm^3 gas/s)
(i)	5
(ii)	12
(iii)	2

SECTION B

11. The rate of decomposition of a solution of hydrogen peroxide is increased by the presence of a catalyst.

(a) (i) Write down an equation for the decomposition. (1½ mark)

(ii) Name a suitable catalyst. (½ mark)

(iii) What term is used for a catalyst that takes part in a chemical reaction in a living organism? Name the catalyst contained in raw liver. (1 mark)

(b) Draw a labeled diagram of an apparatus you could assemble to measure the rate of decomposition of hydrogen peroxide in the presence of manganese (IV) oxide.

Indicate clearly how the reaction could be started at a time known exactly and how the gas produced is collected and its volume measured. (3 ½ marks)

(c) A student as a choice of using the same mass of finely powdered manganese (IV) oxide or coarsely powdered (granulated) manganese dioxide. Which of these would you expect to have a greater average rate of reaction over the first minute of the reaction? Give a reason for your answer. (1½ marks)

(d) A set of results obtained in an experiment to measure the rate of decomposition of hydrogen peroxide, in a solution of known volume and concentration, is given in the table.

Time (minutes)	0	1	2	3	4	5	6	7	8
Volume of oxygen (cm^3)	0.0	13.5	23.4	30.5	35.4	38.3	39.6	40.0	40.0

Plot a graph to illustrate the volume of oxygen produced versus time. (4 mark)

(e) Use the graph to determine:

(i) the volume of oxygen produced during the first 2.5 minutes (1 mark)

(ii) the instaneous rate reaction at 2.5 minutes. (1 mark)

(f) What changes woud you expect in the graph if the experiment was repeated using a solution of the same volume but exactly half the concentration of the original hydrogen peroxide solution? (1 mark)

12. (a) Draw a fully labeled diagram and give an equation to show how you would prepare a dry sample of sulphur dioxide in the laboratory starting from a namd sulphite. (4 marks)

(b) Sulphuric acid is made by the contact process. $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$

This is carried out in the presence of a catalyst at 450^0C and 2 atmospheres pressure.

(i) How is the sulphur dioxide made? Write the equation for the reaction (2 marks)
(ii) Give one other use of sulphur dioxide. (1 mark)
(iii)Name the catalyst used. (1 mark)
(iv)If the temperature is decreased to 300^0C, the yield of sulphu trioxide increases, explain why this lower temperaure is not used. (1 mark)
(v) Sulphur trioxide is dissolved in concetrated sulphuric acid. This is added to water to make more sulphuric acid, why is sulphur trioxide not added directly to water?

(c) 9.12g of an hydrous iron (II) sulphate was heated according to the following equation.

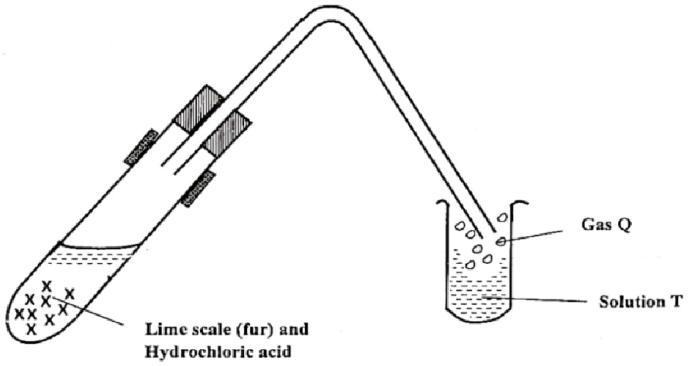
$$2FeSO_4(s) \longrightarrow Fe_2O_3(s) + SO_3(g) + SO_3(g)$$

(i) the mass of iron (III) oxide formed. (1 mark)
(ii) the volume of the gaseous products at room temperaute and pressure. (1 mark)

(d) Give equations only for the reactions which take place when hydrogen sulphide, H_2S, reacts with: (3 marks)
(i) an excess of air (oxygen) (iii) aqueous iron (III) chloride.
(ii) lead (II) nitrate solution.

13. (a) (i) What is hard water? (1 mark)
(ii) What type of water hardness can be removed by boiling, name or give the formula of the substance that can cause this type of water hardness. (2 marks)
(iii) identify one problem caused by hard water. (1 mark)
(iii)Give one advantage of hard water. (1 mark)

(b) The diagram below shos a test tube containing limw scale (fur scraped from the inside of a kettle) and dilute hydrochloric acid.



As the gas X produced by the reaction is bubbled through solution T, the solution turns milky (cloudy). Identify the gas Q and solution T and give the equation for the reaction. (3 marks)

(c) Consider the following chemical and materials used in the treatment and purification of a drinking water supply.

- Sand and gravel, Potash alum (potassium aluminium sulpahte), Carbon slurry, Sodium fluoride, NaF, Lime (CaO) and Chlorine.

(i) What term is used to describe the use of potassium aluminium sulphate causing suspended particles in water to clump together? (1 mark)
(ii) Give the purpose of sand and gravel in water purification. (1 mark)
(iii) What is the purpose of adding chlorine to the water supply. (½ mark)

(iv) Under what circumstance would lime (CaO) and carbon slurry be added to the water supply. (1 mark)

(d) A mass of 1.62 g of calcium hydrogen carbonate was completely converted to calcium carbonate, water and carbon dioxide by the action of heat.

(i) What mass of the residue was produced? (2 marks)

(ii) What volume of acrbon dioxide was produced at standard temperature and pressure? (1 mark)

14(a) When dilute hydrochloric acid is warmed with each of the following compounds, a gas is evolved, name this gas and give a chemical test to establish its identity.

(i) Sodium sulphite. (ii) Calcium carbonate (iii) Iron (II) sulphide.

(b)Washing soda is cheap, household chemical used for laundry, removing grease and softening water. Washing soda crystals are hydrated sodium carbonat ($Na_2CO_3.xH_2O$). The crystals effloresce in dry air becoming powdery in the process.

To determine the average value of x in the formula, $Na_2CO_3.xH_2O$, for a sample of washing soda 3.46 g of the crystals were dissolved in distilled water and made up to exactly 250cm^3 of solution. 25.0cm^3 volume of this solution were pipetted into a conical flask and titrated with a previously standardized hydrochloic acid solution using methyl orange indicator. The balanced equation for the titration reaction is:

$$2HCl\ (aq) + Na_2CO_3\ (aq) \longrightarrow 2NaCl\ (aq) + H_2O\ (l) + CO_2\ (g)$$

On average 21.50cm^3 of 0.12 M hydrochloric acid solution were required to completely neutralise 25cm^3 of the washing soda solution.

Find by calculation:

(i) The average number of moles of hydrochloric acid used. (1 mark)

(ii) The number of moles of sodium carbonate neutralized in each titration. (1 mark)

(iii)The number of moles of sodium carbonate in 250cm^3 of the washing soda solution. (1½marks)

(iv)The mass of sodium carbonate in 250cm^3 of the washing soda solution 1 mark)

(v) The mass of water of crystallization and hence the number of moles of water in 3.46g of crystals. (1 mark)

(vi)The ration of ,oles of water of crystallization of moles of sodium carbonate in the crystals and hence the valve of x to the nearest whole number. (2 marks)

(vii) Explain the meaning of the following terms, giving one example of each: deliquescence and hygroscopic substance. (3 marks)

END

WAKISO-KAMPALA TEACHER'S ASSOCIATION (WAKATA)

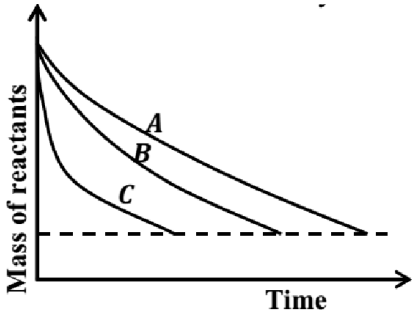
SECTION A

1. (a) A mixture consists of sulphur and Iron fillings. Explain briefly how a sample of sulphur

can be obtained from the mixture. (2½ marks)

(b) A sample of the mixture in (a) was heated in a porcelain dish. State what was observed and write an equation for the reaction that took place. (1½ marks)

2. When hydrogen gas was passed over x g of strongly heated copper (II) oxide until there was no further change, 4 g of a solid was formed.
   (a) State what was observed and write the equation for the reaction. (2½ marks)
   (b) Determine the value of x. (2½ marks)
3. Carbon dioxide was bubbled into a fairly concentrated solution of sodium hydroxide and no visible change was observed at first. On further bubbling, a white precipitate was formed.
   (a) Explain the observation (2 marks) (b) Write an equation (s) for the reaction(s) (3 marks)
4. Ethanol can be prepared by fermentation of sugar.
   (a) Write an equation to show how ethanol can be prepared from glucose $C_6H_{12}O_6$. (1½mks)
   (b) When ethanol is heated with concentrated sulphuric acid at 170^0C, a substance ***W*** is formed.
      (i) Name ***W***. (1 mark) (ii) Write the structural fromula of W. (1 ½ marks)
      (iii) Name one reagent that can be used to identify ***W***. (1 mark)
5. 3.4 g of a mixture containing calcium carbonate and calcium sulphate was reacted with excess hydrochloric acid and 448cm^3 of carbon dioxide measured at s.t.p was evolved.
   (a) Write an equation for the reaction. (1½ marks)
   (b) Calculate:
      (i) the mass of calcium carbonate in the mixture. (2½ marks)
      (ii) the percentage of calcium carbonate in the mixture (1 mark)
6. (a) Name one substance that causes. (2 marks)
      (i) permanent hardness of water. (ii) temporary hardness of water.
   (b) State one advantage of using hard water. (1 mark)
   (c) (i) Write equation to show how permutit(Na_2Y) removes permanent hardness in water(1m
      (ii) State the principle on which permutit works. (1 mark)
7. Element *X* and *Y* with atomic number 12 and 8 respectively react to form a compound *W* .
   (a) Write the electronic configuration of (i) X (ii) Y. (2 marks)
   (b) State whether W is. (2 marks)
      (i) a gas, liquid or solid at room temperature. (ii) convalent or ionic.
8. The graphs below show the effect of temperature on the rate of the reaction between marble clips of the same mass and excess 2M hydrochloric acid



   (a) If curve B is for the reaction of 40^0C, which curve shows the reaction taking place at (2mrks)
      (i) 20^0C (ii) 60^0C
   (b) Explain why the curves eventually end at the same level. (3 marks)

(c) State one other method that can be used to measure the rate of the reaction between marble chips and hydrochloric acid. (1 mark)

9. State what would be observed and write ionic eqiation for the reaction that would take place, if hydrogen chloride was bubbled through aqueous:
   (a) Sodium hydrogen carbonate.(2½ marks) (b) Silver nitrate. (2½ marks)

10. The enthalpy of combustion of carbon is -393kJmol^{-1}.
   (a) Write an equation for the complete combustion of carbon. (1½ marks)
   (b) A bag of 80kg f charcoal costs 40,000/=. Calcualte the cost of charcoal required to produce 16375kJ (C=12, O=16, 1 mol of =12g.) (3½ marks)

SECTION B

11. (a) Describe how you would prepare pure crystals of lead (II) nitrate in the laboratory starting from lead (II) oxide. Write an equation for the reaction that takes place. (7 marks)
   (b) State what happens when lead (II) nitrate is trongly heated. (2 marks)
   (c) State what is observed if ammonia solution is gradually added to a solution of lead (II) nitrate until in excess. Write an equation for the raection that takes place. (3 marks)
   (d) Lead (II) ions raect with iodine ions according to the following equation.

$$Pb^{2+}(aq) + 2I^{-}(aq) \longrightarrow PbI_2(s)$$

   400cm^3 of 0.1M solution of iodide ions was added to a solution containing excess lead (II) ions. Calculate the mass in grams of lead (II) iodide formed. (3 marks)

12. (a) Explain each of the following observations. Write an equation to illustrate your answers.
   (i) When zinc dust is put into a solution of copper (II) sulphate, the blue colour of the solution fades and solution becomes hot. (4 marks)
   (ii) When a test tube filled with chlorine gas is inverted into a trough of water and the setup exposed to sunlight, a gas which relights a glowing splint is produced in the test tube. (3 marks)
   (b) 25.0cm^3 of 0.1M sulphuric acid had a pH less than 7. A solution of sodium hydroxide was gradually added and pH gradually increased. After 20.0cm^3 of the sodium hydroxide solution had been added, the resultant solution had a pH of 7.
   (i) Explin why the pH of the acid increased when sodium hydroxide was added.(3 mrks)
   (ii) Calculate the concentration of the sodium hydroxide solution in moles per litre(5mks)

13. (a) Describe how sulphur is extracted by the Frasch process. (9 marks)
   (b) Write equation(s)to show how fuming sulphuric acid can be obtained from sulphur(5mks)
   (c) State what would be observed if concentrated sulphuric acid is added to sugar. (1 mark)

14. (a) In the laboratory, pure chlorine can be prepared at room temperature by reacting hydrochloric acid with solid **L**. (2 marks)
   (b) Identify L (ii) State the other condition for the reaction
   (iii)Write equation for the reaction that would take place. (1½ marks)
   (b) The gas evolved in (a) (ii) was purified using some substances. (3 marks)
   (i) Name the substances. (ii) Explain the role of the substance named in b (i)
   (c) The chlorine obtained in (b) was passed through aqueous I ron (II) nitrate. (3 marks)

(i) State what was observed. (ii) Explain the observation made in (c) (i)

(d) Aqueous chlorine was exposed to sunlight and a colourless gas M was evolved.

(i) Identify M (ii) State the other observation made. (1½ marsk)

(iii) Write equation for the reaction that would take place. (1½ marks)

END

RESOURCE TEAM MOCK EXAMIATION.

SECTION A.

1. (a) (i) State the approximate percentage of oxygen in the atmosphere. (½ mark)

   (ii)Name one natural process by which oxygen is used up from the atmosphere (½mark)

   (b) Sodium metal was burnt in excess oxygen.

   (i) State what was observed (½ mark)

   (ii) Write an equation for the reaction in (b) (i) above. (1½ marks)

   (c) The product in (b) (i) was dissolved in water. State what was observed and write an equation for the reaction that took place. (2 marks)

2. Amixture of ammonium carbonate and lead(II) chloride was shaken with excess water, filtered and the residue kept.

   (a) Identify the anion that was present in the: (i) filtrate (ii) residue. (1 mark)

   (b) (i)Briefly state how the anion in the residue can be confirmed when present in solution(1)

   (ii) Write ionic equation for the reaction that takes place in (b) (i) above. (1½ marks)

3. 0.22g of anhydrous sodium carbonate reacted completely with excess dilute hydrochloric acid. $50cm^3$ of carbondioxide measured at 800cm temperature and pressure was formed.

   (a) Write an equation for the reaction between sodium carbonate and dilute hydrochloric acid. (1 ½ marks)

   (b) Calculate the: (i) moles of sodium carbonate that reacted. (1½ marks)

   (ii)Moles of carbondioxide evolved. (1 mark)

4. (a) (i) Write an equation for the reaction between Zinc and dilute hydrochloric acid in presence of copper (II) sulphate. (1½ marks)

   (ii) What is the purpose of adding Copper (II) sulpahte in the reaction above? (1 mark)

   (b) Calculate the volume of hydrogen which can be obtained ata s.t.p by heating dilute hydrochlori acid with 5g of Zinc (Zn=65) (2½ marks)

5. (a) Write the structural formula of ethene. (1½ marks)

   (b) (i) Name two compounds that can be used to prepare ethene in the laboratory. (1mark)

   (ii) State the conditions for the reaction in (b) (i) above. (1 mark)

   (c) Ethene was bubbled through bromine water. State what was observed and write the equation for the reaction (2 ½ marks)

6. (a) Element M and N have atomic numbers 6 and 17 respectively.

   Write the electronic confuguration of element N and the ion formed by N. (2 marks)

(b) (i) Write formula of the compound formed between M and N. (1 mark)

(ii) What type of bond exists in the compound formed in b (i) above. Give a reason for your answer (1 mark)

(c) State whether the compound formed between Mand N conducts electricity or not. Give a reason for the answer. (1 mark)

7. When excess carbon monoxide was passed over 4.0g of a strongly heated oxide of an element X, 2.8g X was formed.

(a) Determine the molecular of the oxide X (X=56, RFM of the oxide of X=160) (3½ mark)

(b) Write an equation for the reaction between the oxide of X and carbon monoxide(1½mrks)

8. 2.0g of ammonium nitrate was dissolved in $100cm^3$ of water, and the temperature of the water dropped from 28.0^oC to 22.0^0C.

(a) Give a reason why there was a drop in the temperature of the water. (1 mark)

(b) Calculate the molar enthalpy of solution of ammonium nitrate. (H=1, N=14, 0=16, density of water $=1gcm^{-3}$ and the heat capacity of water$=4.2Jg^{-1o}C^{-1}$) (4 marks)

9. A concentrated solution of sodium chloride was electrolyzed ueing platinum electrodes.

(a) Write ionic equations for the reaction that took place at the electrodes.

(i) Anode. (½ mark) (ii) Cathode. (½ mark)

(b) State what was observed at each electrode. (i) Anode(½ mark) (ii) Cathode. (1½ marks)

(c) State and explain the effect on litmus of the resultant solution formed after the electrolysis of concetrated sodium chloride solution. (1 mark)

10. State the methods used to separate the following (5 marks)

(i) Iron and sulphur

(ii) Sodium chloride and sodium chloride.

(iii) Water and propanol.

(iv) Ammonium chloride and sodium chloride.

(v) Chlorophyl in a green leaf.

x (vi) Air.

SECTION B.

11. (a) What is meant by the following terms: (i) Soap (ii) Saponification. (2 marks)

(b) Outline the steps taken to prepare a dry sample of soap in the laboratory. (no equation is required) (6½ marks)

(c) Briefly explain how soap can remove dirt from your shirt. (3 marks)

(d) When soap is used for washing in hard water, lathering does not occur immediately but when detergent is used for washing in the same water, there is immediate lathering. Expalin this observation. (3½ marks)

12. (a) Define the term rate of chemical reaction. (2 marks)

(b) Explain how the following factors affect the rate of a chemical reaction.

(i) Temperature. (2 marks) (ii) Concentration. (2 marks)

(c) The table below shows variations in the concetration of hydrogen peroxide with time, when hydrogen peroxide reacted to produce oxygen.

Concentration of H_2O_2 ($moldm^{-3}$)	1.15	0.64	0.36	0.24	0.15
Time (Min)	0	10	20	30	40

(i) Plot a graph of concetration of hydrogen peroxide against time. (4 marks)

(ii) Explain briefly the shape of your graph. (2 marks)

(d) Determine from the graph the rate of reaction at: 10 minutes and 30 minutes (3 marks

13. (a) What is meant by the term enthaly of neutralization? (2 marks)

(b) Describe an experiment that can be carried out to determine the enthalpy of neutralisation of hydrochloric acid by sodium hydroxide.(show the working) (6 marks)

(c) When $50cm^3$ of 1M H_2SO_4 acid was added to $50cm^3$ of 2M NaOH, the temperature of the resultant solution rose by 24.6^0C to 38.2^0C.

(i) Write an ionic equation for the reaction that took place. (1½ mark)

(ii) Calculate the enthalpy of neutralization of NaOH. (4½ marks)

(S.H.C of solution $=4.2Jg^{-1}K^{-1}$, density of solution $=1gkm^3$) (1 mark)

(b) Explain why plastic wares are used in neutralization experiments instead of glass wares. (1 mark)

14. (a) Outline with the help of equations how you would prepare a dry sample of hydrogen gas in the laboratory (No diagram is required) (5½ marks)

(b) Draw a labeled diagram of the set up of apparatus that can be used to show that hydrogen gas burns in oxygen to form water. (5 marks)

Calculate the maximum volume of hydrogen measured at s.t.p that can react with 8g of copper (II) oxide.(Cu=64, O=16. 1 mole of a gas occupies $22.4dm^3$ at s.t.p)

$$CuO(s) + H_2\,(g) \longrightarrow Cu\,(s) + H_2O(g)$$ (2 marks)

END

UGANDA MUSLIM TEACHERS' ASSOCIATION (UMTA)

SECTION A

1. (a) Name the method you would use to separate the following mixtures: (4 marks)

(i) Copper (II) carbonate and ammonium chloride. (iii) Kerosene and crude oil.

(ii) Oil and water. (iv) Dissolved salts in sea water.

(b) Copper (II) carbonate was heated until no further change. State what was observed and write the equation for the reaction. (2 ½ marks)

2. (a) Name one allotrope of carbon that is used:

(i) In extration of iron (½ mark) (ii) As an electrode (½ mark)

(b) State one property of the allotrope of carbon that you have named in (a) which is the reason for its use. (2 marks)

(c) Describe how you can prove that the above form of carbon are allotropes of carbon. (2 marks)

3. A mixture of sodium hydroxide solution and ammonium chloride was heated and $180cm^3$ of ammonia gas were evolved s.t.p.

(a) Write an ionic equation for the reaction of sodium hydroxide and ammonium chloride. (1½ marks)

(b) Explain why ammonia can not be collected by:

(i) Downward displacement of water.(1mark) Upward displacement of air. (1 mark)

(c) Calculate the mass of ammonia gas that was produced at s.t.p. (1 mole of a gas occupies 224000cm^3 at s.t.p: H=1 N=14) (1½ marks)

4. Dilute nitric acid was added to a mixture of copper (II) oxide and lead (II) oxide until no further change. To the reslutant solution, dilute sodium hydroxide solution was added drop wise until in excess and the mixture was then filtered.

(a) Write the formula of cation present in: (i) Filtrate. (ii) Residue (1 mark)

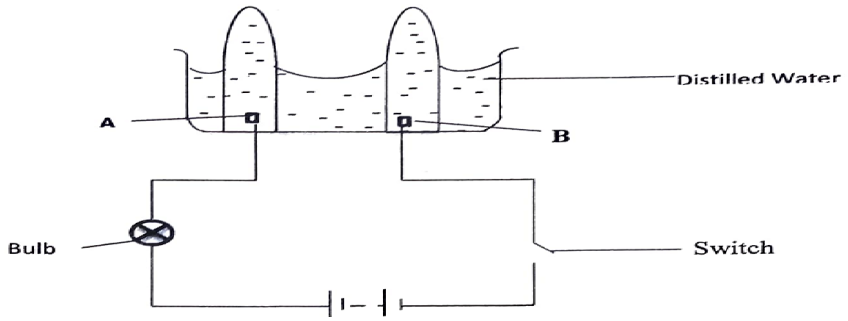
(b) The residue was heated strongly until no further change. State what was observed and write the equation for the reaction. (2½ marks)

(c) In another experiment students had solution oof laed (II) nitrate , magnesium sulphate and zinc chloride.

Which one of the above solution formed a precipitate with: (1½ marks)

(i) Barium nitrate solution. (ii) Silver nitrate solution. (iii) Potasium iodoide solution.

5. Study the diagram below and answer the questions that follow.



(a) Explain why there was no observation made when the switched was closed. (1 mark)

(b) A small amount of concetrated sulphuric acid was added to distilled water and the switch was then closed.

(i) Explain the observation made. (1½ marks)

(ii) Write an equation for the reactions that took place at electrode B.

(c) State one application of electrolysis process. (1 mark)

6. (a) State the conditions under which sulphuric acid can react with: (2 marks)

(i) Copper. (ii) Ethanol.

(b) State the property of sulphuric acid which is shown by its reaction with. (2 marks)

(i) Copper (ii) Ethanol.

(c) Write an equation for the reaction between sulphurica cid and copper. (1½ marks)

7. A hydrated salt contains 20.2% Iron, 23.0% oxygen, 11.5% sulphur and 45.3 water of crystallisation. Its relative molecular mass is 278.

(a) Determine the molecular formula of the hydrated salt.(Fe=56, S=32, O= 16, H=1) (3mks)

(b) 6.95g of hydrated salt were dissolved in distilled water to make 250cm^{3}of solution. Calculate the concentration of the solution in moles per litre. (2 marks)

8. (a) In one experiment, metals A, B and C each were added separately to 50cm^3 of 1M copper (II) sulpahte solution. The initial temeprature of copper (II) sulpahte soltion was noted and recorded before the start of each experiment.

The highest temperature reached of the mixture after stirring was recorded as below.

Metal added to copper (II) sulpahte solution.	A	B	C
Highest temperature (^oC) of the mixture	28.3	24.0	33.7
Initial temperature of copper (II) sulphate solution	24.0	24.0	24.0

(i) Arrange the metals A, B and C in order of reactivity starting with the most reactive and in each case give a reason for your answer. (2 marks)

(ii) Which metal did not show any visible reactio with copper (II) sulpahte. (½ mark)

(iii) State any other observation made inside the reaction vessel in which there was a reaction that occurred. (1 mark)

(b) In the reaction vessels in which there was a reaction.

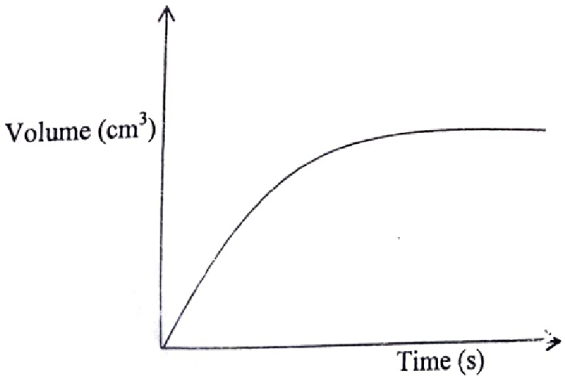
(i) State where the reaction was exothermic or endothermic. Give a reason for your answer. (1 mark)

(ii) State the type of ethalp change in the above experiment. (½ mark)

9. (a) What is meant by the term rate of reaction? (1 mark)

(b) State any two factors that can affect the rate of a reaction. (2 marks)

(c) In an experiment, hydrogen gas was prepared by reacting magnesium ribbon with dilute hydrochloric acid, and the volume of hydrogen gas evolved varied with time, as shown in the graphin the figure below.



(i) On the same axes in (c) above, sketch the curve that would be obtained if a few crystals of copper (II) sulpahte were added into the reactants. (1 mark)

(ii) Explain your answer in (i) above. (1mark)

10. (a) An element W has mass number 27 and 14 neutrons.

(i) Write down the electronic configuration of W. (½ mark)

(ii) W conbines with oxygen to form compound R. write down the formula of R and state the type of bonding in R. (1 mark)

(iii) State any two properties of compound R. (2 marks)

(b) Write an equation for the reaction between R and dilute hydrochloric acid (1½ marks)

SECTION B

11. Glucose can be converted into ethanol by a catalytic reaction caused by the enzyme produced by yeast.

(a) Name (i) the reaction in which yeast converts glucose to ethanol. (1mark)

(ii) the enzyme produced by yeast during the above reaction. (1 mark)

(b) Write the equation for the reaction leading to the formation of ethanol by the process named in a (i) above. (1½ marks)

(c) When etahnol was strongly heated together with concetrated sulphuric acid, gas W was formed: Identify gas W, Write equation for the reaction leading to the formation of gas W. (2½ marks)

(d) (i) Name one reagent that can be used to identify W in the laboratory. (1 mark)

(ii) State what is observed when the reacgent named in d(i) is treated with gas W.(1mark)

(e) When treated at high pressure and temperature in the presence of a suitable catalyst, molecules of W react together to form a compound P of high molecular mass.

(i) Identify p. (½ mark)

(ii) Name the process leading to formation of compound P. (1 mark)

(iii) State any three disadvantages of using P. (3 marks)

(f) Differentiate between thermosetting and thermo-softening plastics. (2 marks)

12. (a) Describe the laboratory methods for preparation of dry sample of sulphur dioxide from sodium sulphite. (Diagram not required) (5 marks)

(b) State what is observed when:

(i) Burning magnesium ribbon is lowered into a gas jar of sulphur dioxide. (1½ marks)

(ii) Sulphur dioxide gas is bubbled through a solution of iron (III) chloride. (1½ marks)

(iii) Write an equation for the reaction that took place in b (i) above. (1½marks)

(c) When sulphurdioxide is reacted with more oxygen in presence of a catalyst, substance X is formed, X dissolves in water forming a solution Z.

(i) Name the catalyst used. (1 mark)

(ii) Identify substance X. (1 mark)

(iii) Write an equation for the formation of X. (1 mark)

(iv) Name solution Z. (1 mark)

(d) Solution Z was reacted with impure zinc to produce hydrogen gas.

(i) Write the equation for the reaction. (1½ marks)

(ii) If 6.5g of impure zinc sample reacted completely with $20cm^3$of a 0.1M solution Z, calculate the percentage of zinc in impure sample at s.t.p (Zn=65) (1½ marks)

13. (a) Differentiate between strong acid and concetrated acid. (2 marks)

(b) The table below shows the pH values for some unknown solution formed.

Solution	pH value.
A	6.2
B	12.0
C	7.0
D	2.0
E	9.8

Which of the above solutions is most likely to be formed when each of the following substances are dissolved in water? Give a reason for your answer. (5 marks)

(i) Carbon dioxide (iv) Hydrogen chloride gas.

(ii) Ammonia gas (v) Sodium chloride.

(iii) Sodium oxide

(c) (i) Calcium oxide is abasic oxide. Define the term basic oxide. (1 mark)

(ii) Describe the laboratory preparation of dry crystals of calcium nitrate from calcium oxide. (5 mark)

(d) Calcium nitrate crystals were strongly heated until no further change. State what was obserbed and write the equation of reaction that took place. (3 marks)

14. (a) Name the main ore from which the following metals are extracted? (2marks)

(i) Sodium metal. (ii) Iron metal

(b) Iron is extracted from its ore in the blast furnace using a reducing agent.

(i) Give the main reducing agent in the blast furnace and write equations for its formation. (2marks)

(ii) Using equations, describe the use of calcium carbonate in the extraction of iron metal from its ore. (3 marks)

(c) State what is observed when iron metal is heatedin dry sample of chlorine gas. (2 marks)

(d) Explain why common reducing agents are not used in the extraction of sodium metal from its ore. (2 marks)

(e) During extraction of sodium, calcium chloride is added to the ore.

(i) State the purpose of adding calcium chloride to the ore. (1 mark)

(ii) Write equations for the reactions at electrode during extraction of sodium metal. (3marks)

END.

UGANDA TEACHERS' EDUCATION CONSULT (UTEC)

SECTION A

1. In an experiment, cooking oil was added into a beaker containing a large volume of water and the mixture was shaken then aalowed to stand.

(a) State what was observed. (½ mark)

(b) (i) Name the piece of apparatus that can be used to separate the mixture. (½ mark)

(ii) Give a reason for your answer in (b) (i) above. (½ mark)

(c) In another experiment, ammonium chloride was heated as a mixture with sodium chloride.

State: (i) What was observed during the process of heating? (1 mark)

(ii) The practical application of heating the mixture. (½ mark)

(d) Mention one example of a mixture which can be separated by fractional distillation (1 mark)

2. The atomic number of elements, X, Y and Z are 6, 12 and 17. Respectively:

(a) Write the electronic configuartion for: X, Y and Z. (1½ marks)

(b) Write the formula for the product when:

(i) X, burns in excess oxygen. (1 mark)

(ii) Atoms of Y are burnt in the molecules of Z. (1 mark)

(c) Water was added to the compound in (b) (i) and the products tested with litmus paper (s). state: what was observed? Any one conclusion thar can be drawn following your observation above. (1½ marks)

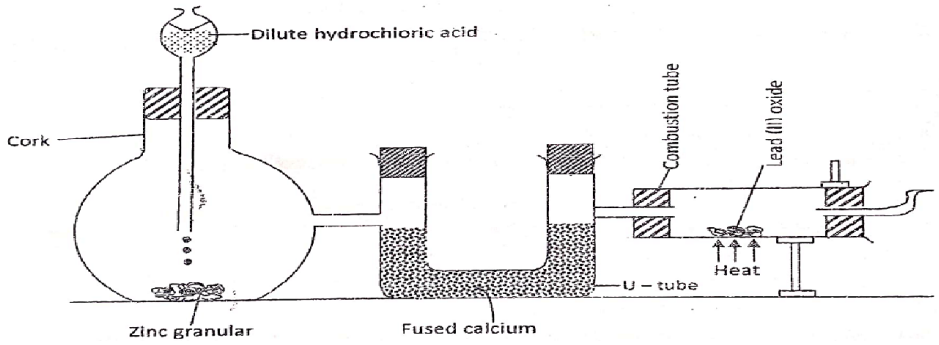
3. (a) Sodium peroxide reacts with water to produce oxygen according to the following equation. $2Na_2O_2(s) + 2H_2O(l) \longrightarrow 4NaOH\ (aq) + O_2(g)$.

(i) State what was observed during the reaction. (1 mark)

(ii) Calculate the volume of oxygen gas produced at room temperature when 3.12g of sodium peroxide were completely reacted with water. (Na=23, O=16, I mole of a gas occupies 24dm^3 at room temperature) (2 marks)

(b) A small sample of burning sulphur was plunged into a gas jar containing oxygen gas. State: What was observed?, the effect of the product(s) on litmus paper. (2 marks)

4. Dilute hydrochloric acid was added until in excess to zinc granules and the gas evolved was passed over heated lead (II) oxide as shown in the figure 1 below.



(a) State:

(i) What was observed in the combustion tube? (1 mark)

(ii) The role of fused calcium chloride in the U-tube. (½ mark)

(b) Write equation for the reaction.

(i) Between zinc and dilute hydrochloric acid. (1½ mark)

(ii) That takes place in the combustion tube. (1½ mark)

(c) Give one other oxide besides Lead (II) oxide that could have been used in the combution tube. (½ mark)

5. A solid compound, Q reacts with dilute sulphuric acid to produce a colourless gas that has the following properties:

(i) Turns acidified potassium dichromate solution from orage to green.

(ii) Bleaches moist blue flower to white.

(a) Name: the compound Q and the colourless gas. (1 mark)

(b) Write an equation for the reaction between Q and dilute sulphuric acid. (1½ mark)

(c) State:

(i) The practical application of the reaction when the gas was bubbled through acidified potassium dichromate solution. (½ mark)

(ii) What would be observed if the colourless gas was passed through a solution of Iron (III) sulphate. (1 mark)

(d) The bleaching action of the colourless gas by reduction. Write equation for the reaction to show how the gas reduces moist blue flower. (1 mark)

6. When magnesium powder was added into $25cm^3$ of 0.2 M solution of copper (II) chloride in a plastic beaker, some heat was evolved.

(a) State: What else was observed apart from the evolution of heat: Why the reaction liberates haet. (2 marks)

(b) Write an ionic equation for the raection that took place. (1 mark)

(c) Calculate the:

(i) Number of moles of copper (II) ions contained in the $25.0cm^3$ of the solution (1 mark)

(ii) The mass of copper that was displaced by one mole of magnesium. If 1.92g of magnesium were used in the reaction (Cu=64, Mg=24) (2 marks)

7. The diagrma in fig.2 below shows the set up of apparatus that was used to carry out electrolysis of differet electrolytes in the laboratory. Carbon electrodes were used in the experiment.

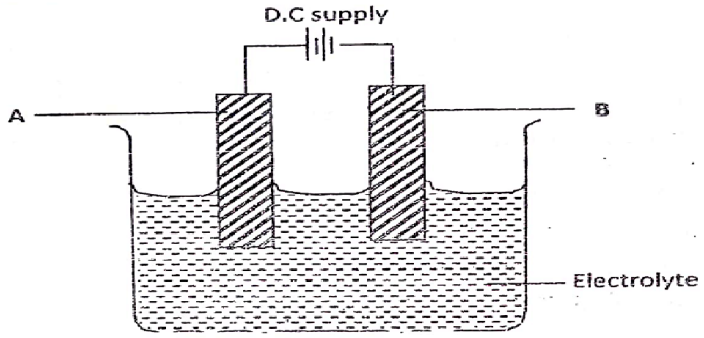


Fig. 2

(a) Name the substance that was formed at telectrode, A, if the following electrolytes were used : Aquesous silver nitrate solution and copper (II)sulphate in dilute form (1mark)

(b) Write an equation for the lectrode reaction for aqueous silver nitrate solution. (1mark)

(c) If the electrolte usedwas concetratec sodium chloride solution, name the product (s) that was formed at electrodes : A and B (1 mark)

(d) State how the product you have given in (c) for electrode B can be identified in the laboratory. (1 mark)

(e) If the electrode were repaced with copper electrodes and the electrolysis in (a) for copper (II) sulpahet in dilute form done for some time, state what was observed at A. (½ mark)

8. (a) State what would be observed if each of the following reagents was added to a solution of aluminium chloride.

(i) Lead (II) nitrate solution (ii) Ammonium solution dropwise until in excess(1½mks)

(b) Write an equation for the reaction that took place in (a) (i) above. (1 mark)

(c) To the resultant mixture in (a) (ii) was added sodium hydroxide solution dropwise until in

excess: State what was observed and give a reason for your answer. (2 marks)

9. (a) In the presence of a catalyst, glucose can be converted to ethanol.
   (i) Name the catalyst. (½ mark)
   (ii) write the eqaution for the reaction in which glucose is converted to ethanol. (1 marks)
   (iii) State how the purity of ethanol can be determine. (½ mark)
   (b) The ethanol obtained can be converted to carbon dioxide gas according to the following Equation. $C_2H_5OH(l)+3O_2(g) \longrightarrow CO_2(g)+H_2O(l) + Heat$
   Calculate the mass of ethanol that was burnt in excess oxygen to produce 544.0KJ of heat (C=12, O=16, H=1: 1 mole of ethanol burns to produce 1360KJ of heat)
   (c) State what:
   (i) is formed when etahnol is heated with concetrated sulphuric acid. (½ mark)
   (ii) is observed when the product in c (i) is treated with bromine liquid. (1mark)

10. When concetrated hydrochloric acid was added to manganese (IV) oxide and the mixture heated, agas was evolved.
   (a) (i) Name the gas that was evolved. (½ marks)
   (ii) Write equation for the reaction that took place. (1½ marks)
   (b) State what was observed when the gas was burbled through:
   (i) Cold dilute potassium hydroxide solution. (1 mark)
   (ii) A beaker containing moist red flowers. (½ marks)
   (c) The gas in (a) (i) was bubbled through a solution of sodium iodide until no further change.
   (i) Write equation for the reaction. (1½ marks)
   (ii) Suggest any conclusion that can be drawn from the equation you have written in (c) (i). (1 mark)

SECTION B

11. Calcium oxide is regarded as a hygroscopic substance.
   (a) (i) Define the term a hygroscopic substance. (1 mark)
   (ii) Give one other example of a hygroscopic substance. (½ mark)
   (iii) State one laboratory use of hygroscopic substance. (1 mark)
   (b) A gas A, is mainly dried in the laboratory using calcium oxide.
   (i) Identify gas A. (½ mark)
   (ii) Write an equation for the reaction that leads to the formation of gas A. (1½ marks)
   (c) Excess water was added to calcium oxide until there was no further change.
   (i) Write equation for the reaction that took place. (1½ marks)
   (ii) Using equations, explain the changes that would result if excess carbon dioxide gas was bubbled through the product(s) in c (i) until in excess. (4½marks)
   (d) State:
   (i) The conditions for the reaction of calcium oxide with nitric acid. Write equation for the reaction that took place. (2½ mark)
   (ii) Why the reaction in (d) (i) is not possible with sulphiric acid (equation not required)

(2 marks)

12. Hydrogen chloride gs can be prepared in the laboratory by reacting sodium chloride crystals and an acid.
    (a) (i) Name the acid commonly used for this preparation. (½ mark)
        (ii) State the conditions for the reaction. (1 mark)
        (iii) Write equations for the reaction. (1½ marks)
    (b) Iron fillings when heated in dry hydrogen chloride react according to the following equation: $Fe(s) + 2HCl(g) \longrightarrow FeCl_2(s) + H_2(g)$
        (i) Calculate the volume of hydrogen chloride gas that would be required to reactwith iron at s.t.p to produce 5.0g of iron (II) chloride. (Fe=56, Cl =35.5: 1 mole of a gas occupies 22.4dm^3 at s.t.p) (2½ marks)
        (ii) The an hydrous iron (II) chloride formed in (b) was dissolved in water. State what was observed and write an ionic equation for the reaction. (2½ marks)
    (c) Dilute sodium hydroxide solution was added to the solution (product) in (b) (ii) drop wise until in excess.
        (i|) State what was observed (1 mark)
        (ii) Write an ionic equation for the reaction. (1½marks)
    (d) Explain:
        (i) The effect of bubbling hydrogen chloride gas in the solution of silver nitrate.(2½mks)
        (ii) Why aqueous hydrogen chloride liberates hydrohen with zinc metal whereas hydrogen chloride in methyl benzene doesn't . (2½ marks)

13. Nitrogen can react with hydrogen in the Haber process to produce ammonia gas.
    (a) State the: (i) source of nitrogen and hydrogen. (1 mark)
        (ii) factors that can affect the yield of ammonia gas in the Haber process (1½ marks)
    (b) Write an equation for the reaction leading to the formation of ammonia in the Harber process. (1½ marks)
    (c) Write an equation for the reaction to show oxidation of ammonia gas. (3 marks)
        (i) By copper (II) oxide (ii) in the presence of hot platinum.
    (d) Using equations, outline how the product(s) in (c) (ii) can be converted to nitric acid. (3 ½ marks)
    (e) Ammonia gas can react with concentrated sulphuric acid to produce ammonium sulphate according to the following equation: $2NH_3(g) + H_2SO_4(l) \longrightarrow (NH_4)_2SO_4(S)$
        (i) Calculate the mass of ammonium sulpahte that would be produced by 0.96dm^3 of ammonia gas at room temperature. (N=14, S=32, 0=16, H=1, onw mole of a gas occupies 24.0dm^3 at room temperature. (2 marks)
        (ii) Explain why an aqueous solution of ammonium sulphate tuurns blue litmus paper red. (2 marks)
    (f) When aqueous ammonia solution was added to a solution containing zinc ions. A white precipitate Y which dissolved in excess ammonia to give a colourless solution was formed. Identify Y ad write the formula for the cation in the colourless solution. (1 mark)

14. (a) State the difference between an acid and a salt. (2 marks)

(b) Describe:

(i) How a pure dry sample of lead (II) carbonate can be prepared in the laboratory. (no diagram required). (2 marks)

(ii) The effect of heat on lead carbonate. (2 marks)

(c) Lead (II) carbonate reacts with dilute nitric acid according to the following equation.

$$PbCO_3(s) + 2HNO_3(aq) \longrightarrow Pb(NO_3)_2(aq) + CO_2(g) + H_2O(l)$$

Calculate the mass of lead (II) carbonate that is required to react completely with 200cm^3 of 0.2 M dilute nitric acid. (Pb=207, C=12, o=16) (2½ marks)

(d) State what be observed if a solution of ;ead (II) ions was added into:

(i) Drops of potassium iodide. (½ marks)

(ii) Ammonia solution drop wise until in excess. (1mark)

(iii)Dilute hydrochloric acid and the mixture heated then allowed to cool. (1½ marks)

(e) Write an equation to illustrate your answr in d (ii) above. (1½ marks)

END

KAYUNGA SECONDARY SCHOOLS EXAMINATION COMMITTEE (KASSEC)

SECTION A

1. In a sugar factory, the sweet juice which is dirty because of soli, green matter, small fibres and water, is treated with calcium oxide: the juice is boiled and lastly passed throug animal charcoal.

(a) Suggest the aim of:

(i) Treating sugar cane juive with calcium oxide. (1 mark)

(ii) Boiling the sugar cane juice. (1 mark)

(iii)Passing the juice throgh animal charcoal. (1 mark)

(b) Cane sugar (sucrose) was treated with concetrated sulphuric acid in a glass beaker.

(i) State what was observed. (1 mark)

(ii) Suggest the property of sulphuric acid shown. (1 mark)

2. A hydrocarbon contain 4.8g of carbon and 1.0g of hydrogen (C=12,H=1)

(a) Determine the empirical formula of the hydrocarbon. (2½ marks)

(b) (i) If the relative formula mass of the hydrocarbon above is 58. Find its molecular Formula (2 marks)

(ii) Write down the structural isomers of the hydrocarbon. (2 marks)

3. X, Y and Z in the table below represent atoms of different elements:

Element	Proton number	Neutron number	Atomic mass
X	6	6	……
Y	8	…….	16
Z	12	……..	24

(a)(i) Fill in the missing values in thhe table above. (1½ marks)

(ii) Write down the electronic configuration of element Y. (½ mark)

(iii)State the group of element X. (½ mark)

(b) Write down the formula of the compound formed between elements Y and Z. (1 mark)

(c) Using outer most energy levels only draw a diagram to show how bonding occurs between X and Y. (1 mark)

4. (a) When a nitrate of a metal Y was heated strongly, brown fumes were observed and a solid residue which has was brown when hot and yellow when cold. Identify , metal Y and write the chemical equation for the reaction that took place. (2½ marks)

(b) Amixture of solid ammonium chloride and solid calcium hydroxide was heated and a pure gas was collected.

(i) Write the equation for the reaction. (1½ marks)

(ii) name the drying agent for the gas produced. (1mark)

(iii)Suggest the method of gas collection and give a reason. (1 mark)

5. When certain equipment is left in most air, it becomes covered with rust material.

(a) Write the chemical formula of rust material. (1 mark)

(b) One of the rust material removers contains hydrochloric acid.

(i) State what is observed when the rust remover is applied on the rust material. (1mark)

(ii) Write the chemical equation for the reaction that occurs in b (i) above. (1½ mark)

(c) (i) Give one disadvantage of rusting. (1 mark)

(ii) Suggest one way of preventing rusting. (1 mark)

6. (a) Define the following terms.

(i) Anode (ii) electrolyte. (2 marks)

(b) A concentrated solution of sodium chloride was electrolyzed using carbon electrodes.

(i) Name the products at each electrode (anode and cathode) (1 mark)

(ii) Write the equation at each of the electrodes. (anode and cathode) (2 marks)

(iii)

7. (a) Define each of the terms below. (3 marks)

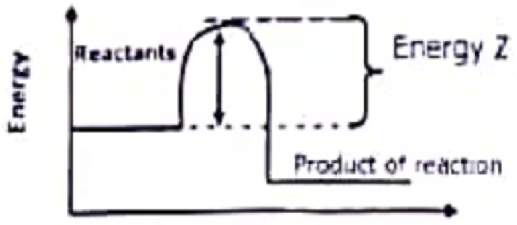
(i) An acid. (ii) PH. (iii) Neutralization.

(b) Complete and balance each of the following equations. (2 marks)

(i) $H_2SO_4(aq) + CuO(s) \longrightarrow$

(ii) $HNO_3(aq) + Mg(s) \longrightarrow$

8. (a) The graph below shows the energy changes for a reaction. Study it to answer the questions that follow.



(i) State whether the reaction is exothermic or endothermic. (½ mark)

(ii) Name energy Z. (1 mark)

(b) When ethanol is burnt, energy is produced. In one experiment, the temperature of 0.1kg of water was raised by 32.6°C by burning o.46g of ethanol. Assuming no heat loss.

(i) Calculate the amount of heat produced by 0.46g of ethanol (C_2H_5OH) (C=12, 0=16, H=1, S.H.C of water=42000j) (2 marks)

(ii) Calculate the heat of combustion of ethanol in $KJmol^{-1}$. (2 marks)

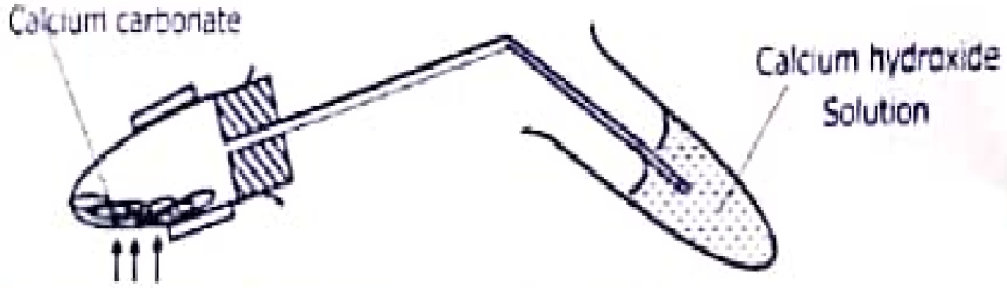
9. Copper reacts with oxygen according to the following equation.

$$2Cu\ (s) + O_2(g) \longrightarrow 2CuO(s)$$

(a) Determine mass of copper oxide that can be obtained from 32g of copper? (2 marks) (Cu=64, O=16)

(b) Determine the volume of oxygen required at room temperature to produce 8g of copper (II) oxide. (Molar gas volume at room temperaure=$24dm^3$) (2 marks)

10. Calcium carbonate was heated strongly in a boiling tube and the gas produced was bubbled through a dilute solution of calcium hydroxide.



(a) State and explain what was observed in the test tube if:

(i) Little gas was passed through calcium hydroxide in test tube. (2½ marks)

(ii) Excess gas was passed through calcium hydroxide solution in the test tube.(2½marks)

(b) Write the equation of the reaction in (a) (i) above. (1½marks)

SECTION B

11. (a) State three factors that affect the rate of a reaction. (3 marks)

(b) Amixture of a known mass of zinc granules and a certain volume of 2M sulphuric acid was put in a conical flask. The volume of the gas produced was recorded at various intervals.

Time (s)	0	10	15	20	25
Volume of gas (cm^3)	0	5	10	15	15

(i) Plot a graph of volume of gas produced against time. (5 marks)

(ii) From the graph, determine the rate at which the gas was produced when time was 5 seconds and 22 seconds. Give a reason for the difference of the rate. (5 marks)

(c) On the same axex sketch graphs that would be obtained when the experiment was repeated with:

(i) 1M sulphuric acid instead of 2M sulphuric acid.

(ii) Powdered zinc instead of Zn granules. (2 marks)

12. (a) (i) Name the process that leads to formation of soap. (1 mark)

(ii) Name two sources of vegetable oils that can be used during preparation of soap(2mks)

(b) Briefly describe how a sample of soap is prepared in the laboratory. (7 marks)

(c) Sate:

(i) One advantage of using a detergent over soap when washing. (1 mark)

(ii) One disadvantage of the detergents to the environment. (1 mark)

(d) (i) Water containing magnesium hydrogen carbonate does not lather readily with soap unless after boiling. Explain. (1 mark)

(ii) water containing calcium sulphate will not lather even after boiling. Explain (1 mark)

13. (a) (i) Describe with the aid of a labeled diagram, how you can prepare a dry sample of chlorine gas in the laboratory. (6 marks)

(ii) Write an equation for the reaction that takes place in (a) (i) above. (1½ marks)

(iii) State two uses of chlorine. (2 marks)

(b) Eith the aid of equations, state what would be observed if chlorine was bubbled through.

(i) Iron (II) chloride solution. (ii) potassium iodide solution. (2 marks)

(c) State how chlorine gas is tested in the laboratory. (1½ marks)

14. (a) (i) Give two examples of metals that can be found from the earth's crust in their most pure state. (2 marks)

(ii) Name the most two reactive metals. (2 marks)

(b) Iron has always been extracted from a variety of ores.

(i) Name two ores of iron. (2 marks)

(ii) Describe how iron can be extracted from one of its ores. (7 marks)
(equations are necessary)

(iii)State how you would distinguish between Fe^{2+} and Fe^{3+} ions. (2 marks)

END

CENTRAL REGION TEACHERS' ASSOCIATION (CERETA)

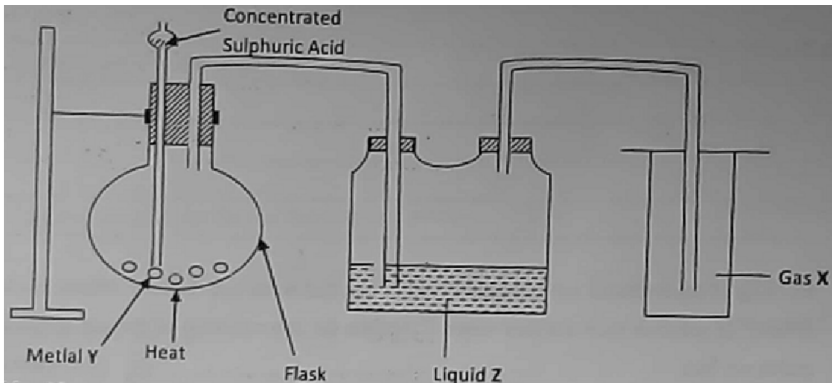
SECTION A:

1. A mass of 25.0 g of copper (II) carbonate was heated strongly until there was no further change.
   (a) State what was observed? (1 mark)
   (b) Write an equation for the reaction. (1½ marks)
   (c) Calculate the mass of the residue. (Cu=65, 0=16, C=12) (2½ marks)
2. Nitric acid reacts with copper to form a colourless gas, which on exposure to air gives brown fumes, soluble in water.
   (a) State the conditions under which nitric acid reacts with copper. (1 mark)
   (b) Name the colourless gas. (½ mark)

(c) Explain how the brown fumes are formed using equation. (2 marks)

(d) Give one use of the colourless gas that relights a glowing splint. (1 mark)

3. The diagram below represents set up of apparatus used in the lab for preparation of gas X



(a) Identify: X, Y and Z. (1½ marks)

(b) Write the equation for the reaction that took place in the flask. (1½ marks)

(c) State what would be observed when gas X is passed through an aqueous solution of acidified potassium permanganate. (1½ marks)

(d) Name the process above. (½ mark)

4. A hydrated salt S has the following composition by mss, Iron 20.2%, sulphur 11.5%, oxygen 23% and water 45.3%, its relative formula mass is 278.

(i) Determine the formula of the hydrated salt S. (3½ marks)

(ii) 13.9g of the hydrated salt were dissolved in distilled water and the total volume made to $500cm^3$ of solution with distilled water. Calculate the concentration of the salt solution in moles perlitre. (1½marks)

5. Vinyl chloride undergoes addition polymrization to form polymer P.

(a) Define the term addition polymerisation. (1 mark)

(b) Write down the structures of Vinyl chloride and P. (2 marks)

(i) Structure of vinyl chloride. (ii) Structure of polymer P.

(c) State what is observed when propene gas is bubbled through a solution of bromine water and write equation for the reaction. (2 marks).

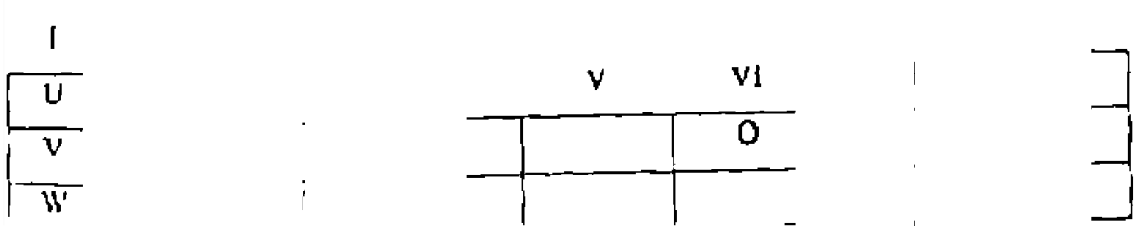
6. Excess hydrogen chloride gas was bubbled through asolution of 10.8g of silver nitrate.

(a) State what was observed. (½ marks)

(b) Write equation for the reaction that took place. (1½ marks)

(c) Calculate the mass of the solid formed. (3 marks)

7. Part of the periodic table is shown below.



(a) Write the formulae of the compounds formed between the following pairs of elements and each case state the type of bonding.

(i) Z and P. (ii) V and O. (2 marks)

(b) (i) Which of the compounds in (a) would conduct electricity in molten state. (1 mark)

(ii) Explain your answer in b (i) above. (1 mark)

(c) State the most:

(i) Electropostive element. (1 mark)

(ii) Inert element. (1 mark)

8. (a) Burning magnesium was plunged into a gas jar of carbon dioxide gas. State what was observed and explain the observation. (2½ marks)

(b) Chlorine gas was bubbled into potassium iodide solution. State what was observed and write an equation for the reaction that took place. (2½ marks)

9. A mixture of sulphur and iron fillings were heated until a reaction began. When the burner was then removed, the reaction mixture continued to glow. A black solid remained.

(a) (i) How would you show that sulphur and iron fillings is a mixture? (1 mark)

(ii) Write an equation for the reaction that took place. (1 mark)

(iii)What does the observation made after removing the burner indicates. (½ mark)

(b) Describe briefly the action of dilute hydrochloric acid on the black solid. (1½ marks)

10. Sulphur(IV) oxide reacts with oxygen in the contact process according to the following equation.

$$2S \quad (\quad \rightleftharpoons \quad 2SO_3$$

(a) State the conditions that increase the yield of sulphur (IV) oxide. (1½ marks)

(b) Write equations of reactions leading to the production of sulphuric acid from sulphur (VI) OXIDE. (3 marks)

(c) State one use of sulphuric acid. (1 mark)

SECTION B

11. (a) (i) State two properties which show that air is a mixture. (2 marks)

(ii) Name two other gases other than oxygen that are constituents of air and give their approximate percentages in air. (2 marks)

(c) Describe an experiment to determine the percentage of oxygen in air. Show how the percentage can be calculated from the results. (8½ marks)

(d) (i) State obervation, if burning sodium is lowered into a jar of oxygen. (1½ marks)

(ii) Write the name and formula of the product of the reaction between iron and oxygen in the above reaction. (1 mark)

12. (a) What is meant by the following terms:

(i) Soap (ii) Saponification (2 marks)

(b) Outline the steps taken to prepare a dry sample of soap in the laboratory. (no equation is required) (6½ marks)

(c) Briefly explain how soap can remove dirt from your shirt. (3 marks)

(d) When soap is used for washing in hard water, lathering does not occur immediately but when detergent is used for washing in the same water, there is immediate lathering. Expalin this observation. (3½ marks)

13. (a) Define the term rate of chemical reaction. (2 marks)

(b) Explain how the following factors affect the rate of a chemical reaction.

(i) Temperature. (ii) Concentration.

(c) The table below shows variations in the concetration of hydrogen peroxide with time, when hydrogen peroxide reacted to produce oxygen.

Concentration of H_2O_2 ($moldm^{-3}$)	1.10	0.60	0.32	0.20	0.10
Time (Min)	0	10	20	30	40

(i) Plot a graph of concetration of hydrogen peroxide against time. (4 marks)

(ii) Explain briefly the shape of your graph. (2 marks)

(d) Determine from the graph the rate of reaction at: 15 minutes and 05 minutes (3 marks

14. (a) What is meant by the term enthaly of neutralization? (2 marks)

(b) Describe an experiment that can be carried out to determine the enthalpy of neutralisation of hydrochloric acid by sodium hydroxide.(show the working)(6 marks)

(c) When $50cm^3$ of 1M H_2SO_4 acid was added to $50cm^3$ of 2M NaOH, the temperature of the resultant solution rose by 24.6^0C to 38.2^0C.

(i) Write an ionic equation for the reaction that took place. (1½ mark)

(ii) Calculate the enthalpy of neutralization of NaOH. (4½ marks)

(S.H.C of solution $=4.2Jg^{-1}K^{-1}$, density of solution $=1gkm^3$) (1 mark)

(e) Explain why plastic wares are used in neutralization experiments instead of glass wares. (1 mark)

END

KANUNGU DISTRICT JOINT MOCK EXAMINATIONS

SECTION A.

1. (a) Explain why solid particles are alwys vibratoing in their positions under room temperature. (1 ½ marks)

(b) Explain what would happen to the solid particles if the temperatures were increased. (1½ marks)

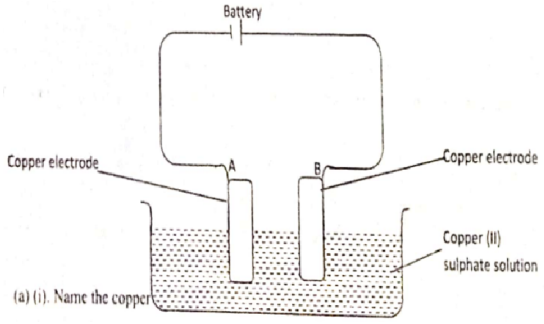
(c) How are the particles in solid different from those in liquid state. (2 marks)

2. (a) An atom of element Q has 12 neutrons and mass number 23. Write the formula of the oxide of Q. (1 mark)

(b) The oxide of Q was dissolved in water and the aqueous solution tested with litmus solution. State what was observed and write equation for the reaction between the oxide of Q and water. (2 marks)

(c) A piece of Q was ignited and lowered into gas jar containing chlorine. State what was observed and write equation for the reaction that took place. (2 marks)

3. Oxygen gas can be best prepared in the laboratory by decomposition of hydrogen peroxide using a catalyst.
   (a) Name the catalyst used and write the requation for the reaction (2 marks)
   (b) Copper metal was burnt in excess air. State what was observed and write the equation for the reaction. (2½ marks)
   (c) The product in (b) was dissolved in warm sulphuric acid, state what was observed(1mark)
4. A hydrated salt P consists of 20% iron, 11.5% sulphur, 23% oxygen and 45.3% water of crystallization.
   (a) Calculate the empirical formula of P. (Fe=56, S=32, O=16, H=1) (2½ marks)
   (b) Determine the molecular formula of P. (Relative formula mass of P=278) (1½ marks)
   (c) Write equation for the reaction between a solution of P and Chlorine. (1 ½ marks)
5. (a) (i) Name the chemical substance that is used to prepare Carbon dioxide in the laboratory with dilute hydrochloric acid. (½ mark)
   (ii) Write an ionic equation for the reaction. (1½ marks)
   (b) Calculate the volume of carbondioxide that would be obtained at s.t.p by dissolving 15g of calcium carbonate in dilute hydrochloric acid. (2 marks)
6. (a) Name the reagent used to differentiate between copper (II) ions and zinc (II) ions(½mark)
   (b) State what is obseerved when the reagent is used on the two cations. (3 marks)
7. (a) Ethene can be prepared from Ethanol.
   (i) Name the type of reaction for the preaparation of ethene from Ethanol. (½ marks)
   (ii) State the conditions for the reaction. (1 mark)
   (iii)Write the equation for the reaction. (1½ marks)
   (b) (i) Name the reagent used to differentiate bewteen Ethene and Ethane. (½ marks)
   (ii) State what is observed when the reagent is used. (1 mark)
8. Copper (II) sulpahte solution was electrolysed using copper electrodes as shown below.



   (a) (i) Name the copper electrodes A and B. (1 mark)
   (ii) Name the ions present in the electrolyte. (2 marks)
   (b) State what is observed at (i) electrode A (ii) electrode B. (1 mark)
   (c) Write an equation for the reaction that takes place on each electrode:
   Electrode A and electrode B (3 marks)
9. Iron can be extracted from one of its ore Haematite.

(a) (i) Define an ore.(1 mark) (ii) Write the chemical formula of Haematite ore. (½ mark)

(b) In the extraction of iron ore Haematite coke and limestone are fed into a blast-furnance and hot air is blown into the mixture.

(i) Why is lime stone added to the mixture? (1 mark)

(ii) Write equations for the reactions that lead to the formation of iron. (2 marks)

10. (a) What do you understand by Enthalpy of combustion. (1 mark)

(b) When 0.382 g of Ethanol in a specimen bottle were burnt completely heat produced made the teperature of water in a thin-walled tin rise from 17.7^0C to 41.2^0C. Mass of water in the tin can is 100g: specific heat capacity of water is $4.2Jg^{-1o}C^{-1}$. (3½ marks)

SECTION B.

11. (a) (i) Draw a alabelled diagram of the set up of apparatus that can be used to prepare a dry sample of ammonia in the laboratory. (3½ marks)

(ii) Write equation for the formation of ammonia. (1½ marks)

(b) (i) Name the process by which ammonia is manufactured on large scale. (½ mark)

(ii) State the conditions for the reaction for the manufacture of Ammonia gas.(1½ marks)

(iii)Write the equation for its manufacture. (1½ marks)

(c) (i) State what is observed when concentrated nitric acid is added into a boiling tube containing copper turnings. (1½ marks)

(ii) Write an equation for the reaction. (1½ marks)

(d) (i) State what is observed when ammonia solution is heated with copper(II) oxide(1mark)

(ii) Write an equation for the reaction. (1½ marks)

12. (a) (i) Name the acid used to prepare chlorine gas in the laboratory with potassium manganate (VII). (½ mark)

(ii) Write an equation for the reaction. (1½ marks)

(b) State the conditions under which iron can react with hydrochloric acid and write equation for the reaction. (2 ½ marks)

(c) Draw a fully labeled diagram for the set up of apparatus ehich can be used to prepare an hydrous iron (III) chloride in the laboratory. (4 marks)

(d) During the manufacture of chlorine commercially, graphite anode and Mercury cathode electrodes are used.

(i) Name the process by which chlorine is produced commercially. (½ mark)

(ii) Write the equation at the cathode and at the anode. (3 marks)

(iii)Name the other products produced. (1 mark)

(e) (i) Write an equation for the reaction between magnesium bromide solution and chlorine.

(ii) Name the type of reaction above in c(i) (½ mark)

13. (a) Name: (1½ marks)

(i) The chemical method used to prepare copper (II) sulphate crystals in a laboratory.

(ii) The chemical susbtances used to prepare copper (II) sulphate crystals in the laboratory

(b) Describe with an equation how you can prepare copper (II) sulpahte crystals in the laboratory using the chemicals named in a (ii) above. (5 ½ marks)

(c) (i) Describe how you can prepare sulphurdioxide gas in laboratory (no diagram required) (3 ½ marks)

(ii) Describe the bleaching action of sulphurdioxide. (2 ½ arks)

(iii)State any four uses of sulphuric acid. (2 marks)

14. (a) (i) What do you understand by rate of a chemical reaction? (1 mark)

(ii) State the factors that affect the rate of a chemical reaction. (2 marks)

(b) Describe an experiment that you would use to determine the rate of a reaction bewteen magnesium metal and hydrochloric acid in the laboratory. (4 ½ marks)

(c) During an experiment for determination of the rate of reaction between magnesium metal and dilute hydrochloric acid the results obtained were recorded as shown below.

Time (s)	0	10	20	30	50	60	70	80
Volume (cm^3)	0	18	30	40	53	57	58	58

(i) Plot a graph of thevolume of the gas producrd against time in seconds. (6 marks)

(ii) Determine the rate of reaction from the graph. (1½ marks)

END.

JINJA JOINT EXAMINATIONS BOARD-JJEB

SECTION A.

1. When heated , ammonium chloride undergoes both physical and chemical change.
   (a) State what would be observed when a physical change occurred.
   (b) Name the process which took place in (a)
   (c) Name one other substance which when heated undergoes the named process in (b).
   (d) Write equation for the reaction that would take place when a chemical change occurs.
2. (a) State what would be observed if litmus was added to each of the following aqueous substance (2 marks)
   (i) Lemon juice. (ii) sodium carbonate (iii) Ammonium sulphate. (iv) sodium chloride.
   (ii) Account for the observations in (a) (ii) (3 marks)
3. (a) State the difference between hard water and soft water.
   (b) State what would be observed if the following aqueous solution **xxxx**
   (i) Magnesium sulphate. (ii) Calcium hydrogencarbonate.
   (c) Write equation for the reaction that took place in (b). (1½ marks)
   (d) State the applicationof the reaction in (b) (ii) (1 mark)
4. Carbon dioxide can be prepared in the laboratory by reacting calcium carbonate with a suitable dilute acid V.
   (a) Identify V. (1 mark)
   (b) Write ionic equation for the reaction leading to formation of carbon dioxide (1½marks)
   (c) When excess carbon dioxide was passed through concentrated sodium hydroxide, a white solid W was formed. (2 ½ marks)
   (i) Identify W (ii) Write equation for the reaction that took place.

5. When sucrose, $C_{12}H_{22}O_{11}$, was reacted with sulphuric acid, a black solid Q was formed.
   (a) Name solid Q (1 mark)
   (b) (i) State the conditions for the reaction to take place (1 mark)
   (ii) Write equation for the reaction leading to formation of Q. (1 mark)
   (c) State the property shown by sulphuric acid. (1 mark)
   (d) Name one other substance other than sucrose which when reacted with sulphuric acid under the same conditions produces Q. (1 mark)
6. The number of protons in atoms X, Y and Z are 13, 6 and 11 respectively
   (a) Write the electronic configuration of atom: X, Y and Z (3 marks)
   (b) When X and Y each react with oxygen separately, they form an oxide. Write the formula of the oxide formed by X and Y. (2 marks)
   (c) Name the two classes of oxides formed when Z combines with oxygen (1 mark)
7. $20.0cm^3$ of 0.5M copper (II) sulphate solution was added to excess sodium hydroxide solution and the mixture shaken.
   (a) State what was observed (½ marks)
   (b) Write equation for the reaction that took place. (1½ marks)
   (c) Calculate the mass of he solid product formed. (2½ marks)
8. When excess dry ammonia was passed over 0.80g of a heated oxide of Iron, 0.56g of Iron was formed.
   (a) Calculate the formula of the oxide. (2½ marks)
   (b) State the chemical name of the oxide. (1 mark)
   (c) Write equation for the reaction that took place. (1½ marks)
9. (a) Aqueous copper (II) sulphate was electrolyzed between grahite electrodes.
   (i) State what would be observed at the cathode. (½ mark)
   (ii) Write equation for the reaction that would take place at the cathode. (1½marks)
   (iii)State the application of the set-up. (½ mark)
   (b) The electrolysis in (a) was repeated using copper anode.
   (i) State what would be observed at the anode. (½ mark)
   (ii) Write equation for the reaction that would take place in b (i) (1½ marks)
   (iii)State the application of this set-up. (½ mark)
10. Excess aqueous sodium chloride was added to $25.0cm^3$ of 0.1M silver nitrate solutin.
   (a) State what was observed. (½ mark)
   (b) Write ionic equation for the reaction that would take place. (1½ marks)
   (c) Calculate the mass of the solid formed. (2½ marks)
   (d) State the practical application of the reaction. (½ marks)

SECTION B.

11. When heated strongly copper (II) nitrate gives a black solid G.
   (a) (i) State the other observations made during the decomposition. (1½ marks)
   (ii) Write equation for the reaction leading to formation of G. (1½ marks)

(iii) Identify G (½ mark)

(b) Starting with G describe how pure crystals of a salt can be prepared using sulphuric acid. (5 marks)

(c) State what would be observed if to the aqueous solution of the salt prepared in (b) was added excess.

(i) Sodium hydroxide solution (1 mark)

(ii) Ammonia solution. (2 marks)

(iii) Write ionic equation for the reaction that would take place in (c) (i) (1½marks)

(iv) Write the chemical name and formula of the compound in the final product in (c) (ii) (1½ marks)

12. Sodium is extracted from its ore by electrolysis where the ore is mixed with calcium chloride.

(a) (i) State what is meant by the term **"ORE"** (1 mark)

(ii) Write the name and formula of the ore from which sodium is extracted. (2marks)

(iii) State why sodium is extracted by electrolysis but not reduction of its oxides. (1 mark)

(iv) State the role of calcium chloride in the extraction of sodium. (1 mark)

(b) (i) Name the substance that is used as the anode and cathode respectively during the extraction of sodium. (2 marks)

(ii) Name the electrode at which sodium is formed and describe the reaction that would take place. (3 marks)

(iii) State one use of sodium. (½ mark)

(c) (i) Name the bi-products obtained during the extraction of sodium from its ore. (1mark)

(ii) State the industrial use of the bi-products obtained during the extraction of sodium from its ore. (1 mark)

(d) When sodium reacts with oxygen, it forms two different products. Write equations showing how sodium reacts with oxygen. (3 marks)

13. (a) In the laboratory, pure chlorine can be prepared at room temperature by reacting hydrochloric acid with solid **L**. (2 marks)

(i) Identify **L** (ii) State the other condition for the reaction

(iv) Write equation for the reaction that would take place. (1½ marks)

(c) The gas evolved in (a) (ii) was purified using some substances. (3 marks)

(i) Name the substances. (ii) Explain the role of the substance named in b (i)

(c) The chlorine obtained in (b) was passed through aqueous Iron (II) nitrate. (3 marks)

(i) State what was observed. (ii) Explain the observation made in (c) (i)

(d) Aqueous chlorine was exposed to sunlight and a colourless gas **M** was evolved.

(i) Identify **M** (ii) State the other observation made. (1½ marsk)

(iii) Write equation for the reaction that would take place. (1½ marks)

14. (a) When sulphuric acid reacts with sodium chloride, two types of salts are formed.

(i) State what is meant by the term "salt" (1 mark)

(ii) Write the name and formula of each of the salts formed. (2 marks)

(ii) Name the class of salts to which each of the salts named in (a) (ii) belong. (2 marks)

(c) During the preparation of lead (II) nitrate starting with lead (II) carbonate, the salt was heated to obtain compound R which was later reacted with a dilute acid T to form Lead (II) nitrate.

(i) Identify the substance **R** and **T**. (2 marks)

(ii) Write the equations for the reactions leading to formation of R and Lead (II) nitrate (3 marks)

(d) Whe heated, excess sulphuric acid reacs with etahnol to form gas E.

(i) Identify the subtance E. (1mark)

(ii) State the conditions for the reaction. (2 marks)

(iii)Write equation for the reaction that would take place. (1½ marks)

(iv)State one application of sulphuric acid. (½ mark)

END

BULO PARENTS S.S.

SECTION A.

1. (a) State a suitable method by which pure sodium chloride can be obtained from its mixture with (1½ marks)

   (i) Iron fillings (ii) Potassium carbonate. (iii) Iron (III) chloride.

   (b) Tetrachloromethane does not dissolve in water whereas ethanol mixes easily with water at all proportions. (3 marks)

   (i) What is observed if 250cm^3 of water is separately shaken with 10cm^3 of ethanol.

   (ii) What is observed if 250cm^3 of water is separately shaken with 10cm^3 of tetrachloromethane.

   (iii)How a mixture of water and tetrachloromethane can be separated.

2. (a) Write the name and formula of one salt that causes permanent hardness of water. (1mark)

   (b) State one physical and one chemicalmethod of removing permanent hardness in water.

   (i) physical (ii) Chemical (1 mark)

   (c) Write equation for the reaction that takes place during removal of permanent hardness from water by the chemical method in (b) above. (1½ marks)

   (d) State one advantage and one disadvantage of using hard water. (1 mark)

3. (a) During the laboratory preparation of hydrogen at room temperature zinc reacted with sulphuric acid or hydrochloric acid but not nitric acid or ethanoic acid.

   (i) Write an ionic equation for the reaction leading to formation of hydrogen. (1½ marks)

   (ii) State the condition for the reaction in a (i) above. (½ mark)

   (iii)State the method of collecting hydrogen gas. (½ mark)

   (iv)Give the reason why laboratory preparation ogf hydrogen cannot be done using:

   Nitric acid: Ethanoic acid. (1 mark)

4. The table below shows the respective number of protons, neutrons and electrons in particles: O, P, Q, R, S and T.

Particles	Protons	Neutrons	Electrons
O	11	12	11
P	17	18	17
Q	12	12	12
R	10	10	10
S	6	6	6
T	13	14	10
U	7	8	7

(a) Identify which particle(s) (2marks)

(i) belong to a metalloid element (ii) belong to the most reactive metal.

(iii) form an amphoteric oxide. (iv) react with cold water to form acidic solution.

(b) Write the formula of compound that would be formed between each of the following pairs of particles and in each case state the type of bonding.

(i) P and S. (1 mark) (ii) Q and U (1 mark)

5. (a) When hydrogen chloride was bubbled through a solution containing a certain cation Q, a white shinny and crystalline precipitate was formed. The precipitate dissolve when the mixture was heated, and recrystallized on cooling the solution. (2 marks)

(i) State the identity of Q.

(ii) Write an ionic for the reaction that took place between hydrogen chloride and Q.

(b) Silver nitrate reacts with sodium chloride to form silver chloride according to the following equation. $Ag^{+}(aq) + Cl^{-}(aq) \longrightarrow AgCl\ (s)$.

Calculate the maximum mass of silver chloride that would be formed if excess sodium chloride solution was addeded to $20cm^3$ of 0.5M silver nitrate solution. (Ag=108, Cl=35.5) (2 ½ marks)

6. (a) Ethanol, C_2H_5OH undergoes dehydration to form a gas W.

(i) Name one common laboratory reagent that can cause the dehydration of ethanol(½mk)

(ii) State the condition(s) for the reaction between the reagent in (a) (i) above and ethanol to form W. (1 mark)

(iii)Write equation to show formation of W. (1 mark)

(b) A liquid R was formed when bromine solution in tetrachloromethane was added to W.

(i) Name R. (½ mark) (ii) State the appearance of R. (½ mark)

(c) Write equation for the complete combustion of W. (1½ marks)

7. Warm dilute nitric acid was added to a mixture of lead(II) oxide and copper (II) oxide and the solution divided into two portions.

(a) To the first portion was added dilute sodium hydroxide dropwise until in excess and the mixture filtered. Identify the cation in the. (i) Filtrate (ii) Residue) (2 marks)

(iii) Write equation for the reaction that lead to formation of the residue. (1½ mark)

(b) To the second portion was added aqueous ammonia dropwise until in excess and filtered. State the colour of the residue and the filtrate. (2 marks)

(c) Write the formula of the cation in the filtrate. (1 mark)

8. During the laboratory prepation of carbon dioxide, calcium carbonate reacts with dilute hydrochloric acid according to the following equation.
$CaCO_3(s) + 2HCl(aq) \longrightarrow CaCl_2(aq) + CO_2(g) + H_2O(l)$.
   (a) Calculate the maximum volume of carbon dioxide in cm^3 that would be produced at room temperature if dilute hydrochloric acid reacted completely with4.5g of calcium carbonate. (C=12,0=16, Ca=40, Imole of gas occupies 24.0dm^3 at room temperature. (3 marks)
   (b) A quantity of dilute sulphuric having the same hydrogen ion concentration as that of hydrochloric acid in (a) above was reacted with 4.5 g of calcium carbonate at room temperature.
      (i) State how the maxmum volume of carbon dioxide would compare with your answer in (a) (1 mark)
      (ii) Give a reason for your answer in (b) (i) above. (1 mark)
9. An aqueous solution of copper (II) sulpahte was electrolysed between carbon electrodes.
   (a) State what was observed at the (i) anode. (ii) cathode (2 marks)
   (b) Writeequation for the reaction that took place at the: (3 marks)
      (i) Anode. (ii) Cathode
10. Diamod and graphite are crystalline allotropes of carbon
   (a) State: (i) What is meant by the term allotrop. (1 mark)
      (ii) How diamond crystal structure differs from agraphite crystal structure. (1 mark)
      (iii) One use of diamond. (1 mark). (iv) one use of graphite. (1mark)
   (b) Name one amorphous allotrope of carbon and state one use of it. (2 marks)

SECTION B.

11. (a) Explain the reactions that lead to the formation of hydrated sodium carbonate crystals. $Na_2CO_3.10H_2O$. from sodium hydroxide solution. (8 marks)
   (c) State what would be observed and in each case, write an equation for the reaction that would take place if.
      (i) Hydrated sodium carbonate crystals were left to stand in air for some time. (2 marks)
      (ii) Sodium carbonate solution was added to copper (II) sulpahte solution. (2½ marks)
      (iii)Anhydrous sodium carbonate is added to excess diulte hydrochloric acid. (2½ marks)
12. (a) Magnetite is one of the ores from which iron can be extracted. During the process, a hot blast of air is fed into the furnace from the bottom. The roasted iron ore and two other substances are introduced in the furnace from the top.
      (i) Write the chemical name and formula of magnetite. (1 mark)
      (ii) Name the two other substances with which the roasted ore is introduced into the furnace. (1 mark)
      (iii)Explain the use of hot air in the furnace. (4 marks)
   (b) Describe with the aid of equations, how in the blast furnace:
      (i) Iron is obtained from magnetite. (2½ marks)
      (ii) The major impurity in the ore is removed. (4 marks)
   (c) Steel is an alloy of Iron.

(i) Nmae one other component of steel other than iron. (½ mark)

(ii) Give two reasons why steel is more used in the construction industry than iron(1mk)

13. (a) (i) Name the method by which zinc sulpahte crystals are prepared from zinc oxide in the laboratory. (1 mark)

(ii) Name the other reagent you can use to prepare zinc sulphate together with zinc oxide. (1 mark)

(iii)Write equation of reaction for the reaction that will occur. (1½ marks)

(iv)Briefly describe how you can prepare zinc sulphate crystals from zinc oxide in the laboratory. (6½ marks)

(v) State one other method by which zinc sulphate crystals can be prepared in the laboratory. (1 mark)

(b) State what is observed when:

(i) Zinc sulphate crystals are added to water stirred. (1½ marks)

(ii) To a portion of the product in (b) (i) is added a few drops of ammonia solution.(1mk)

(c) Write equation of reaction for the reaction occuring in b (ii) above. (1½ marks)

14. (a) Ethene can undergo polymerization. (1½ marks)

(i) Eplain what is meant by the term polymerization. (1 mark)

(ii) Name the product of polymerization of ethene and write equation for the reaction leading to formation of the product you have named. (2½ mark)

(iii)State one use of the product you have name in (a) (ii) (1 mark)

(b) On polymerization, ethene forms a compound, Y, with a molecular mass 16,600.

(i) Determine the number of moles of ethene molecules that combined to form Y. (C=12, H=1) (2 ½ mark).

(ii) State the term which is used to describe a sigle unit of the ethene molecule in Y (1 mark)

(c) (i) Distinguish between the terms synthetic and natural polymer. Give one example in each case.

(ii) State one use of each of the examples in (c) (i) above. (1 mark)

(d) State: (i) one characteristic property of thermoplstic and thermosets. (2 marks)

(ii) one example of a thermoplastic and a thermoset. (2 marks)

END.